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MGT 273 — Marketing Analytics with Power BI

Datasets Used

1. **train.xlsx —** Retail sales data with 9,800 records covering orders, customers, products, regions, and sales.
2. **credits.csv —** Movie cast and crew data with 45,476 records.

These datasets belong to **two different domains** and are used as **separate data sources** in the same Power BI project, each creating its own dashboard page to demonstrate handling multiple data sources.

Section A: Data Preparation (10 Marks)

Importing Datasets (2 Marks)

- Imported train.xlsx via Get Data → Excel Workbook.

- Imported credits.csv via Get Data → Text/CSV.

- Both queries loaded into Power Query Editor for transformation before being applied to the model.

Handling Missing Values, Removing Duplicates, Renaming Columns (4 Marks)

train table:

- Removed duplicate **Row ID** records to make sure each order is counted only once.
- Filled missing values in columns like City, State, Region, Segment, Category, Sub-Category, and Product Name with **"Unknown"** so reports don't show blanks.
- Assigned correct data types (Dates, Decimal Numbers, Whole Numbers) to support accurate calculations and time-based analysis.
- Renamed columns (e.g., *Order Date → OrderDate, Sub-Category → SubCategory*) to make them simpler and more consistent for DAX formulas.

credits table:

- Removed duplicate **id** records so each movie appears only once in the dataset.
- Replaced missing values in **cast** and **crew** with empty lists (**[ ]**) to avoid errors during data processing.
- Renamed columns (e.g., *id → MovieID, cast → castJSON, crew → crewJSON*) to clearly show that these fields contain JSON data.

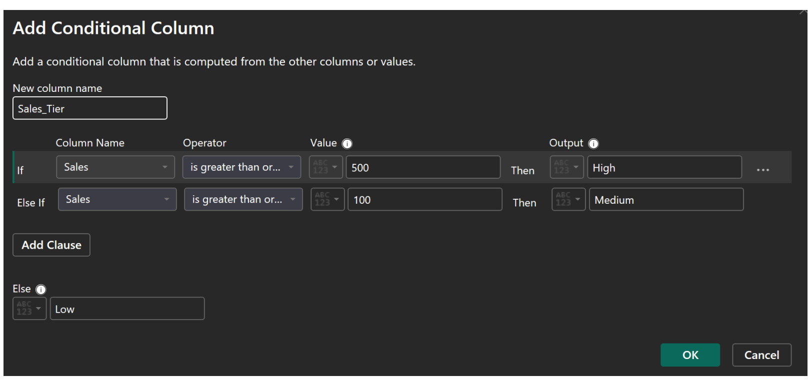
Why these steps matter:
These cleaning steps help keep the data accurate and reliable. Removing duplicates prevents incorrect totals, handling missing values avoids errors, correct data types enable proper calculations, and clear column names make analysis easier to understand.

Creating 2 New Columns — Conditional & Custom

train table

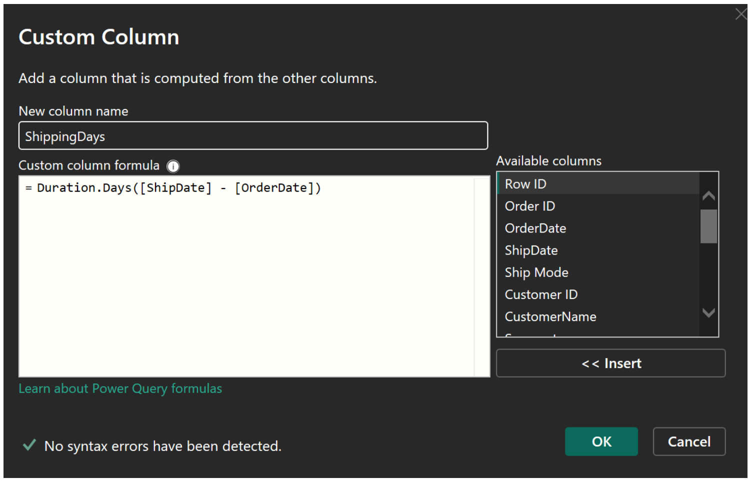
Conditional Column — Sales_Tier

- **Logic:**
  - Sales ≥ 500 → **High**
  - Sales ≥ 100 → **Medium**
  - Otherwise → **Low**
- **Why:**
  This column groups sales into simple categories, making it easier to identify high-value and low-value transactions. It helps create visuals like the Pie chart and supports insights such as finding top-performing and underperforming segments. This type of grouping is commonly used in marketing to plan targeted strategies.

Custom Column — ShippingDays

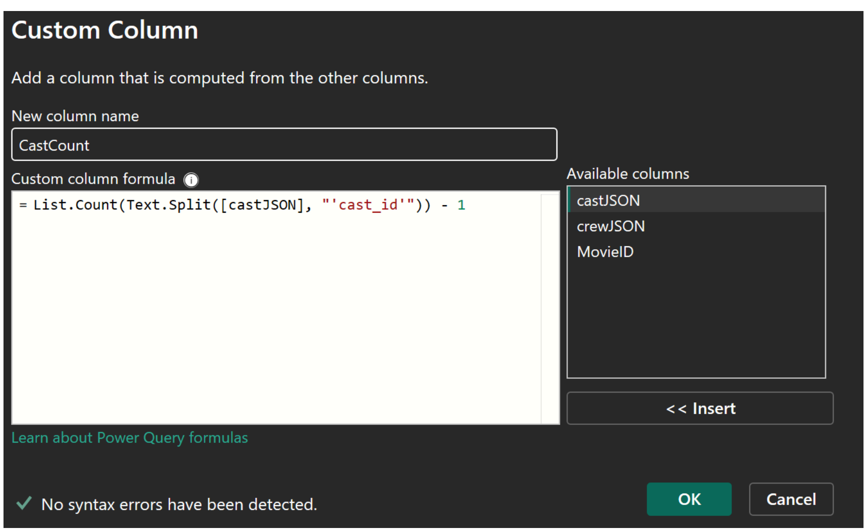
- **Formula:** `Duration.Days([ShipDate] - [OrderDate])`
- **Why:**
  This column calculates how many days it takes to ship an order. Shipping time is an important performance measure in retail operations. It helps analyze delivery efficiency and supports recommendations to improve customer satisfaction and logistics performance.



credits table

Custom Column — CastCount

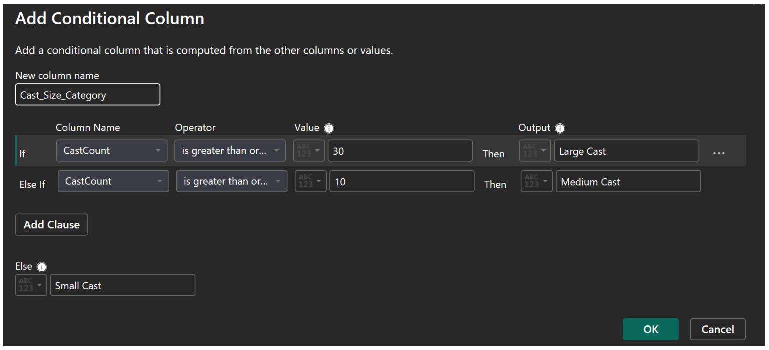
- **Formula:** `List.Count(Text.Split([castJSON], "'cast_id'")) - 1`
- **Why:**
  The original **castJSON** data is in text format and cannot be used directly in charts. This column counts the number of cast members in each movie, turning unstructured data into a useful numeric value that can be analyzed and visualized.



Conditional Column — Cast_Size_Category

- **Logic:**
  - CastCount ≥ 30 → **Large Cast**
  - CastCount 10–29 → **Medium Cast**

- CastCount < 10 → **Small Cast**
- **Why:**
  This column groups movies based on cast size. It makes it easier to compare different types of movies and identify which categories dominate the dataset, supporting analysis of top and underperforming segments in the dashboard.



One more extra Column for both the datasets
train table — Shipping_Speed (Conditional Column)
Pairs with your existing ShippingDays. Logic:
- ShippingDays ≤ 3 → Fast
- ShippingDays 4–6 → Standard
- ShippingDays > 6 → Slow

Why: Turns raw shipping days into a categorical KPI. Useful for slicers, a Pie/Bar visual on delivery performance, and supports the "underperforming segment" insight in Section D.

credits table — CrewCount (Custom Column)
Mirrors your CastCount. Formula:
List.Count(Text.Split([crewJSON], """credit_id""")) - 1

Why: Unlocks crew-side analysis (total production team size). Combined with CastCount, lets you compute a TotalTeamSize measure later and compare cast-heavy vs crew-heavy productions.

Section B: Data Modeling (10 Marks)

1. Create relationships (4 Marks)
The two datasets — train (retail orders) and credits (movie cast & crew) — belong to two completely different business domains with no shared key, no overlapping entity, and no logical join condition.The train table identifies records by Order ID / Customer ID, while credits identifies records by MovieID. There is no column, code, or attribute that meaningfully links a retail order to a movie.Forcing a relationship (e.g., through a synthetic date table or an artificial bridge) would be modeling malpractice — it would create misleading filter contexts and produce numerically valid but logically meaningless results.

The correct modeling decision is therefore to keep the two tables independent in the model. Each dataset drives its own dashboard page, demonstrating Power BI's ability to handle multiple unrelated data sources within a single report.
Relationships created (within each table's own context):

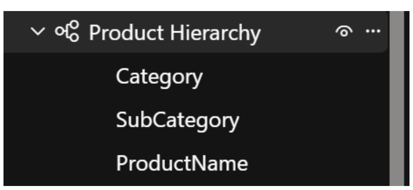
Within the train table: logical referential integrity already exists between Order ID, Customer ID,Product ID, and Region — these act as natural dimensional keys feeding the visuals.
Within the credits table: MovieID is the unique identifier; castJSON and crewJSON are dependent attributes of that key.

Manage Relationships pane: No cross-table relationship is created between train and credits. This is an intentional, documented modeling choice — not a missing step.

2. Create hierarchy (2 Marks)

Train table — Product Hierarchy
Steps:
1. Data view → expand train.
2. Right-click Category → Create hierarchy → rename to Product Hierarchy.
3. Add SubCategory and Product Name to it.

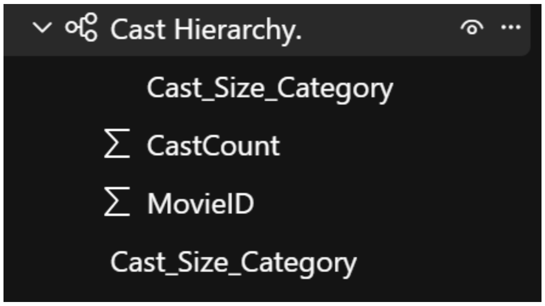
Hierarchy: Category → SubCategory → Product Name
Why: Enables drill-down from Furniture → Chairs → individual product, making retail visuals Interactive.



credits table — Cast Hierarchy

Steps:
1. Data view → expand credits.
2. Right-click Cast_Size_Category → Create hierarchy → rename to Cast Hierarchy.
3. Add CastCount and MovieID to it.

Hierarchy: Cast_Size_Category → CastCount → MovieID
Why: Enables drill-down from cast-size band (Large/Medium/Small) → exact cast count → individual movie, supporting movie-level analysis in Section D.



3. Create calculated column (2 Marks)

Train table — Profit_Estimate

Profit_Estimate = ROUND ('abhinav teja mariyala _ AP23110010233 - train'[Sales] * 0.18, 2)

credits table — Production_Scale

Production_Scale =
IF ('abhinav teja mariyala _ AP23110010233 - credits'[CastCount] + 'abhinav teja mariyala _ AP23110010233 - credits'[CrewCount] >= 100,
"Blockbuster",
IF ('abhinav teja mariyala _ AP23110010233 - credits'[CastCount] + 'abhinav teja mariyala _ AP23110010233 - credits'[CrewCount] >= 30,
"Mid-Scale",
"Indie"))

Why: Combines cast and crew totals into a single production-size classifier. Adds a stronger analytical dimension than Cast_Size_Category alone — used directly in the movie dashboard.

4. Create 2 DAX measures (2 Marks)

4. DAX Measures

The rubric requires 2 measures. To support the full dashboard, additional measures were created and used across KPI cards, drill-through pages, and tooltips. Below are the two primary measures per dataset, followed by the supporting measures.

train table — Primary:

Total Sales = SUM (train[Sales])
Purpose: Core KPI used in cards, line chart, bar chart, donut, and table.

```
Average Order Value =
DIVIDE (
    SUM ( train[Sales] ),
    DISTINCTCOUNT ( train[Order ID] )
)
```

Purpose: Reveals per-order spending behavior; supports top/underperforming segment analysis.

train table — Supporting measures used in dashboard:

Total Orders = DISTINCTCOUNT (train[Order ID])
Avg Shipping Days = AVERAGE (train[ShippingDays])
Purpose: Power the Total Orders and Avg Shipping Days KPI cards plus the Product Detail drill-through page.

credits table — Primary:

Total Movies = DISTINCTCOUNT (credits[MovieID])
Purpose: Drives the Total Movies KPI card and Donut chart.

Average Cast Size = AVERAGE (credits[CastCount])
Purpose: Powers the line chart comparing average cast size across Production_Scale levels.

credits table — Supporting measures used in dashboard:

Avg Crew Size = AVERAGE (credits[CrewCount])
Total Cast Members = SUM (credits[CastCount])
Total Team Size = SUM (credits[CastCount]) + SUM (credits[CrewCount])
Purpose: Power the Avg Crew Size and Total Cast Members KPI cards on Page 2, and the Total Team Size card on the Movie Details drill-through page.

Why so many measures: Each KPI card and detail-page visual needs a dedicated measure rather than a raw aggregation. This keeps the model clean, ensures consistent formatting (currency, decimals, percentages) across visuals, and makes the DAX logic reusable.

Section C: Dashboard Development (15 Marks)

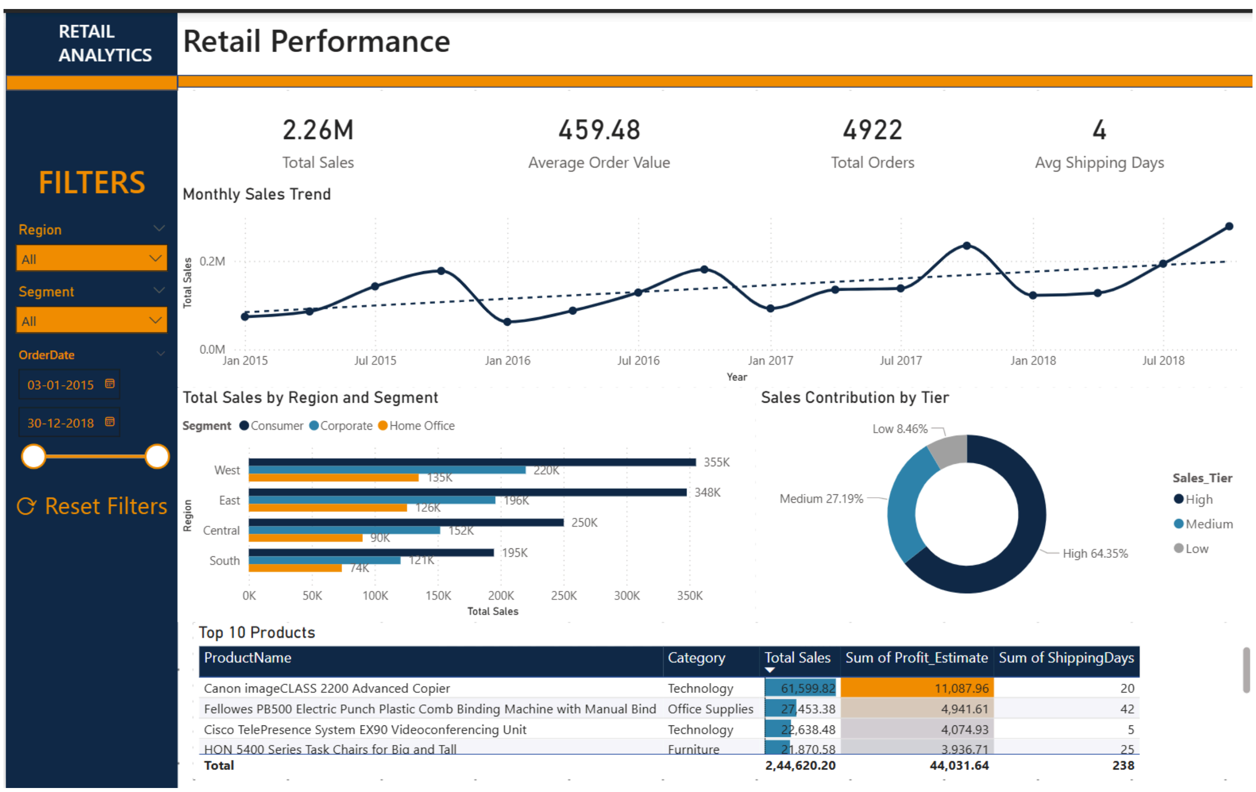
The report contains two independent dashboard pages, each in a consistent "Executive Sidebar" Style with a left navigation rail (brand block,

filters, reset button) and a clean white content area on the right. Both pages share the same fonts (Segoe UI), card design, and layout grid
for visual coherence.

Page 1 — Retail Performance (train dataset)

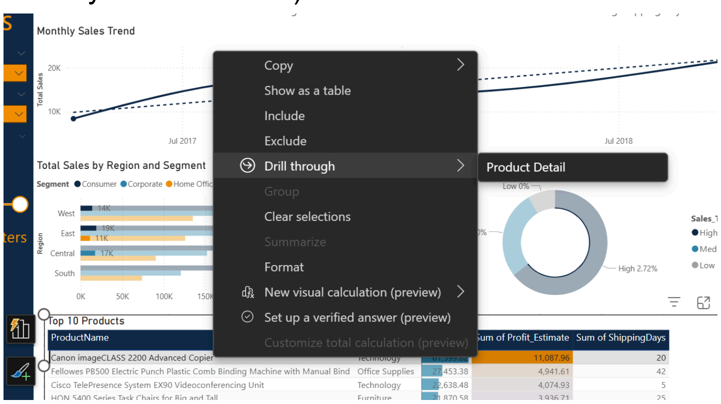
- Brand block: "RETAIL ANALYTICS".
- Filters: Region, Segment, OrderDate (slider).
- Reset Filters button.

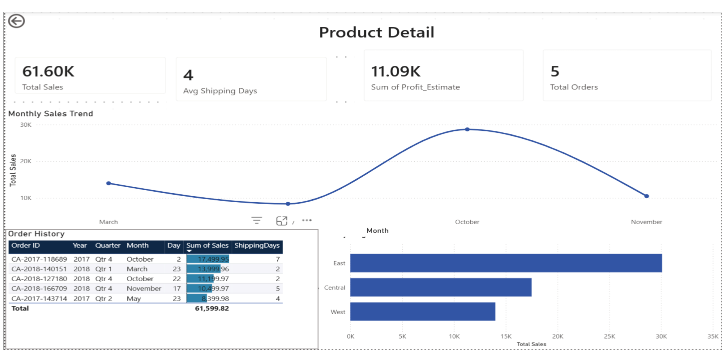
Visuals (5):
1. KPI Cards (4): Total Sales (2.26M), Average Order Value (459.48), Total Orders (4,922), Avg Shipping Days (4).
2. Line Chart — Monthly Sales Trend with dashed trend line showing overall upward growth (Jan 2015 – Dec 2018).
3. Bar Chart — Total Sales by Region & Segment, color-coded by Consumer / Corporate / Home Office.
4. Donut Chart — Sales Contribution by Tier showing High 64.35%, Medium 27.19%, Low 8.46%.
5. Table — Top 10 Products with conditional formatting: blue data bars on Total Sales, orange gradient on Profit_Estimate.



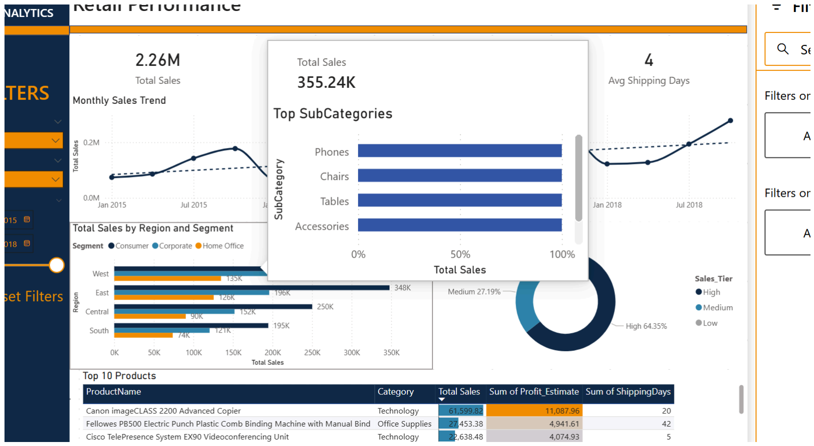
Slicers: Region, Segment, OrderDate.

Drill-through: Right-click any product → Product Detail page (filtered to that product, shows monthly trend, regional sales, full order history with data bars).





Tooltip: Hover the bar chart → Tip_RegionSales page shows Total Sales card + Top SubCategories mini bar.

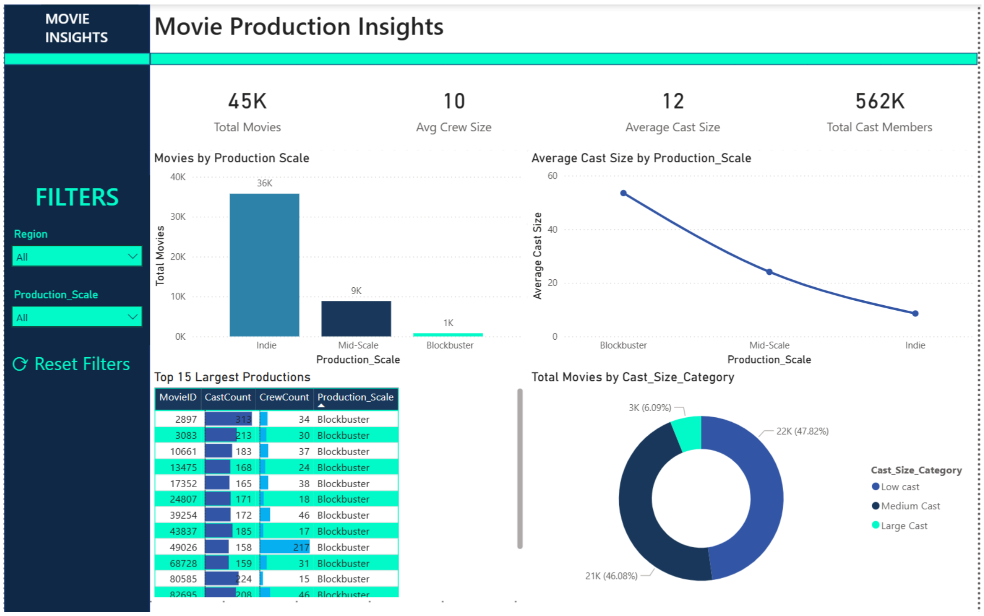


Page 2 — Movie Production Insights (credits dataset)

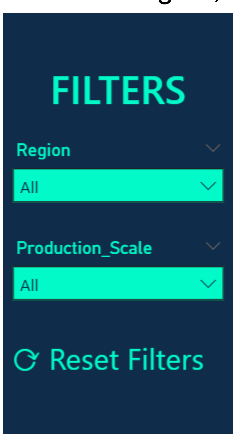
- Brand block: "MOVIE INSIGHTS".
- Filters: Region, Production_Scale.
- Reset Filters button.

Visuals (5):

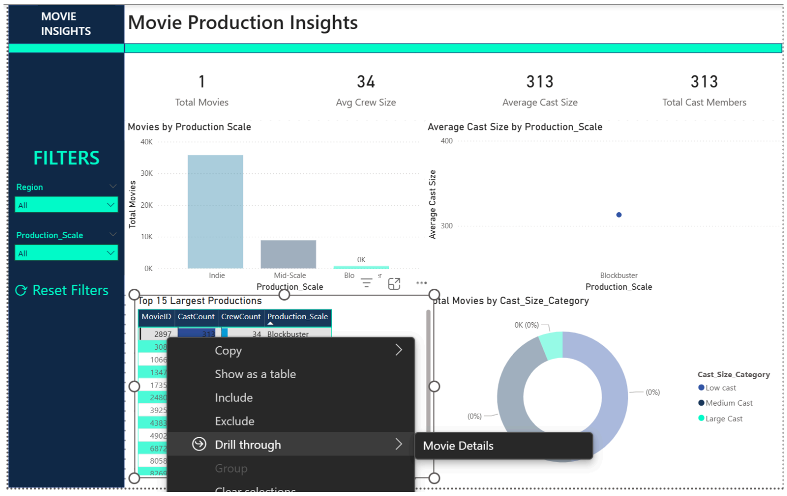
1. KPI Cards (4): Total Movies (45K), Avg Crew Size (10), Average Cast Size (12), Total Cast Members (562K).
2. Bar Chart — Movies by Production Scale: Indie 36K, Mid-Scale 9K, Blockbuster 1K.
3. Line Chart — Average Cast Size by Production_Scale showing Blockbuster (~55) → Mid-Scale (~25) → Indie (~10) decline.
4. Donut Chart — Total Movies by Cast_Size_Category: Low cast 47.82%, Medium Cast 46.08%, Large Cast 6.09%.
5. Table — Top 15 Largest Productions with MovieID, CastCount, CrewCount, Production_Scale and dual-color data bars.

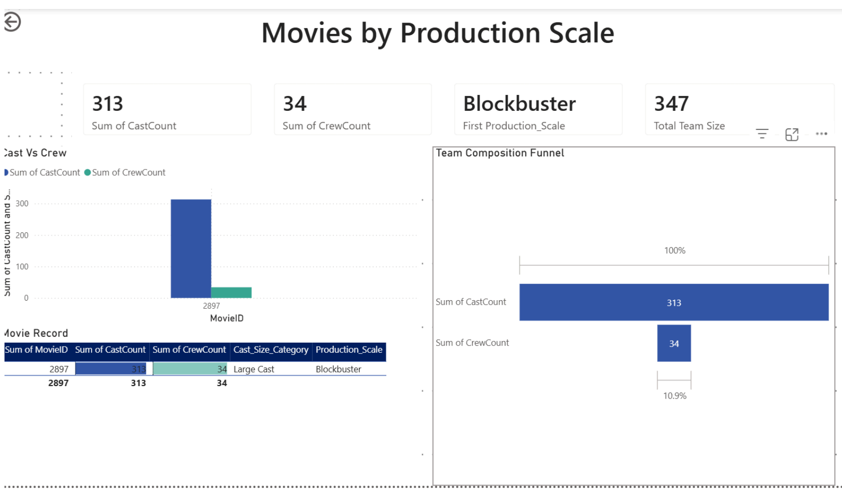


Slicers: Region, Production_Scale.

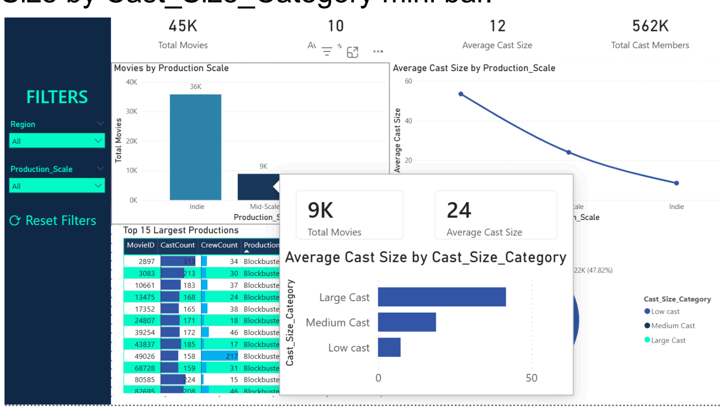


Drill-through: Right-click any MovieID → Movie Details page (shows that movie's CastCount, CrewCount, Production_Scale, Total Team Size, a
Cast vs Crew bar, Team Composition Funnel, and Movie Record table).





Tooltip: Hover bar chart → Tip_MovieScale page shows Total Movies card + Average Cast Size by Cast_Size_Category mini bar.



Section D: Analytical Insights (10 Marks)

1. Top-Performing Segment (2 Marks)

- **Retail Dataset**
  - The West region is the top-performing region with $355K in total sales, followed closely by East ($348K).
  - Consumer customers dominate sales across every region (West Consumer alone ≈ $220K).
  - The High Sales Tier (orders ≥ $500) captures 64.35% of total revenue, confirming that a small group of large orders drives most of the business.
- **Movie Dataset**
  - Blockbuster productions lead in production scale with the highest average cast size (~55 members), significantly above Mid-Scale (~25) and Indie (~10).
  - Blockbusters (1K of 45K total movies) account for the majority of Total Cast Members.

2. Trend Analysis (2 Marks)

- **Retail Dataset (Monthly Sales Trend)**
  - The line chart (Jan 2015 – Dec 2018) shows a clear upward growth trajectory, with sales approximately doubling from ~$0.05M in early 2015 to ~$0.3M by late 2018.
  - The trend reveals strong seasonality:
    - Sharp peaks in November–December every year (year-end demand).
    - Visible dips in January–February (post-holiday slowdown).
    - Sustained year-on-year (YoY) growth across all four years.
- **Movie Dataset (Average Cast Size)**
  - The Average Cast Size by Production Scale line shows a steep declining curve from Blockbuster → Mid-Scale → Indie, indicating that production budgets and team sizes scale exponentially rather than linearly.

3. Underperforming Segment (2 Marks)

- **Retail Dataset**
  - The South region is the weakest performer, with $195K in sales (roughly 45% lower than West).
  - Home Office consistently lags Consumer and Corporate in every region.
  - The Low Sales Tier (orders < $100) contributes only 8.46% of total revenue, indicating that low-value transactions add noise without meaningfully growing the business.
- **Movie Dataset**
  - Indie productions dominate by volume (36K of 45K movies, ~80%) but generate

the smallest cast sizes (avg ~10), indicating a fragmented long-tail with limited scale.
    - The Low Cast category accounts for 47.82% of all movies but contributes minimal cast resources.

4. Business Recommendations (4 Marks)

Based on the patterns above, the following actionable recommendations are proposed:

1. **Double down on the West & East regions** — they already deliver ~62% of total retail sales. Allocate higher marketing spend, expand Consumer-segment campaigns, and prioritize inventory in these regions to compound the existing momentum.
2. **Investigate and revive the South region** — sales are 45% lower than the leading region. Conduct root-cause analysis on shipping efficiency, product availability, pricing competitiveness, and local demand. Consider region-specific promotions and faster fulfillment to lift performance.
3. **Run targeted campaigns during seasonal dips (Jan–Feb)** — the line chart confirms a yearly slump in early months. Introduce off-season offers, bundle deals, or loyalty programs to smooth revenue and reduce dependence on the November–December peak.
4. **Re-examine the Low-Tier and Home Office segments** — these contribute marginal revenue. Either rationalize the product mix to reduce low-margin orders or design upsell campaigns to convert Low-Tier transactions into Medium-Tier ones (≥ $100), where most of the volume opportunity lies.
5. **Balance the movie portfolio** — recognize that Indie productions drive volume but Blockbusters drive scale. Studios should continue funding Indie projects for breadth and audience reach, while reserving larger cast/crew investments for the few Blockbuster bets that deliver disproportionate visibility and team scale.

Section E: Publishing & Storytelling

This Power BI project demonstrates how a single business intelligence solution can be designed to handle two completely unrelated data domains within a single, cohesive interactive report. The two datasets—*train.xlsx* (retail order records) and *credits.csv* (movie cast and crew records)—share no common key, no overlapping entities, and no logical join condition. Rather than forcing an artificial relationship between them, the model intentionally keeps both tables independent, with each driving its own dashboard page. This decision reflects a critical data-modeling principle: a model is judged by correctness, not by the number of relationships drawn between tables.**Retail Performance Dashboard Story**

The Retail Performance dashboard tells a clear growth story:

- **Key Metrics:** Total sales reached $2.26M across 4,922 unique orders, with an average order value of $459 and an average shipping time of 4 days.

- **Top Segments:** The West and East regions together generate roughly 62% of all retail sales, with Consumer-segment customers consistently dominating in every region.
- **Sales Tiers:** The High tier (orders ≥ $500) captures 64.35% of revenue, confirming that a small group of large transactions drives most of the business. The Low tier contributes only 8.46%.
- **Trends:** The monthly sales trend chart reveals consistent year-on-year growth from 2015 to 2018, with a strong seasonal pattern peaking in November–December and dipping in January–February.
- **Underperformer:** The South region stands out, generating just $195K—roughly 45% below the West.

Movie Production Insights Dashboard Story

The Movie Production Insights dashboard tells a very different scaling story:

- The dataset includes 45K total movies and 562K total cast members, with an average cast size of 12 and crew size of 10.
- The data reveals a sharp asymmetry: **Indie** productions account for 80% of all movies (36K films) yet have the smallest average cast size (~10), while **Blockbusters** number just 1K films but command the largest teams (~55 cast members on average).
- The Cast Size Category donut reinforces this, with Low and Medium Cast films collectively representing 94% of the catalog, leaving Large Cast films as a small but disproportionately resource-heavy segment.

Design & Usability Features

Beyond the numbers, the report emphasizes clarity and usability through consistent design and interactivity:

- **Consistent Layout:** Both pages share a consistent **Executive Sidebar** layout (navy left rail with brand block, filters, and a reset button).
- **Distinct Theming:** The pages use distinct accent colors (**orange** for retail, **teal** for movies) so users instantly recognize the domain they are viewing.
- **Interactivity & Navigation:**
  - **Slicers** allow live filtering on Region, Segment, Production_Scale, and Cast Size Category.
  - **Drill-through pages** (Product Detail and Movie Details) let users right-click any item in the top-products or top-productions tables and jump into a fully filtered detail view.
  - **Tooltip pages** (Tip_RegionSales and Tip_MovieScale) appear on hover, providing quick contextual breakdowns without forcing the user to navigate away.
- **Conditional Formatting:** Applied to tables to turn plain data into instantly readable visual stories (e.g., data bars on Total Sales and CastCount, gradient backgrounds on Profit_Estimate, and dual-color bars on Cast vs Crew).

Conclusion

In conclusion, this project showcases not only technical Power BI skills—data cleaning, calculated columns, DAX measures, hierarchies, drill-through navigation, and consistent theming—but also the analytical maturity to recognize that two distinct datasets can coexist meaningfully without being artificially connected. The final dashboards transform raw multi-domain data into clear, actionable insights, providing both operational visibility (sales, shipping, top products) and strategic insight (regional growth opportunities, cast/crew portfolio composition) for decision-makers in retail and entertainment.